Psychoacoustic variables for temporal manipulation of the Sound Induced Illusory Flash

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Abstract

Psychological research on cross-modal auditory visual perception has focused on the influence of visual information on sensory information. There is a lacuna in using an auditory stimulus to manipulate other sensory information. The Sound Induced Illusory flash is one illusory paradigm that uses the auditory system to bias other sensory information. However, more research is needed into the different conditions under which the Sound Induced Illusory Flash manifests and is enhanced or reduced.

The experiment reported here investigates the effect of new auditory variables on the Sound Induced Illusory Flash. The variables to be discussed include the use of varying timescales, pitch intervals, and harmonic relationships. The ultimate aim is to develop the illusory effect as a basis for new multi-media techniques and creative applications for the temporal manipulation and spatialisation of visual objects.

Introduction

Under certain conditions, particular combinations of stimuli – auditory and/or visual - can give rise to non-veridical perception or an illusion. For example in the Müller-Lyer illusion, the horizontal lines are physically equal but the line with the outward arrows at the end is generally perceived to be longer (figure 1.)

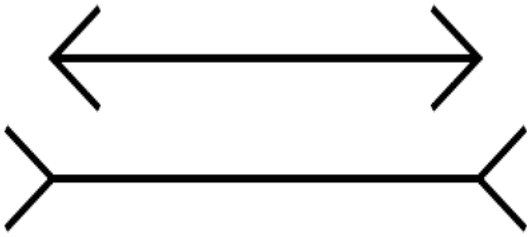


Figure 1. The *Müller-Lyer illusion illustrating that although the lines are the same length, the line with the outward arrows is often perceived as longer.*

This paper investigates non-veridical perception where auditory stimuli influences visual perception. Research on cross-modal interactions in auditory and visual perception have focused predominantly on conditions where visual stimuli are used to manipulate auditory perception (McGurk and Macdonald, 1976) (Alais and Burr, 2004). In such experiments, results suggest that vision is the dominant sense. However, the Sound Induced Illusory Flash illusion (Shams, Kamitani and Shimojo, 2002) exploits the capacity of the auditory system to distort visual perception.

Conditions that give rise to this cross-modal illusion involve presentation of a visual stimulus consisting of a single white dot that is flashed once in the participant's peripheral visual field. This is accompanied by an auditory stimulus consisting of multiple beeps of sound. The participant's task is to judge the number of times the dot is presented. The dual presentation of auditory and visual stimuli appears to emanate

from a single source creating confusion about the number of physical flashes presented and gives rise to the percept of the white dot flashing equivalent to the number of auditory beeps. For example, the dot presented once with three beeps is perceived as three presentations of the dot. This illusory percept appears to occur because of the superior resolution of the auditory system for rhythmic perception, which in this case overrides visual information.

The Sound Induced Illusory Flash illusion is a recent discovery (Shams, Kamitani and Shimojo, 2000) with limited associated research. Whilst recent research (e.g. Shams, Iwaki, Chawla and Bhattacharya, 2005; Shams, 2005; Shams, Kamitani, and Shimojo, 2002) has focused on the underlying neural activity during the illusion, only the initial studies (Shams, 2002) (Shimojo and Shams, 2001) (Shimojo, Scheier, Nijhawan, Shams, Kamitani, and Watanabe, 2001) explored basic structural variables and their effects on the illusion. It is these structural, especially acoustic, variables that are of interest for further exploration of the illusion experience. Such research outcomes provide new opportunities for creative application and multimedia transmission techniques where auditory stimuli might influence visual perception.

Auditory Stimulus Variables and the Sound Induced Illusory Flash – A Review

Rhythm

The variables manipulated in previous research consist primarily of various combinations of the number of auditory and visual stimuli presented (Shams, Kamitani, and Shimojo, 2002), with only minor adjustments to stimuli between experiments (including pitch frequency at 1 kHz or 3.5 kHz; the transmission of auditory stimuli through headphones or speakers; and background screen colour of grey or black). These minute alterations of the stimuli were not considered by Shams and colleagues as significant variables, hence there was little discussion of their effects or interactions.

Brief discussion of the visual stimulus suggested that the illusory percept is stronger when the visual dots are placed in the periphery of the visual field rather than fovea (Shimojo, Scheier, Nijhawan, Shams, Kamitani, and Watanabe, 2001). However, this variable has not been developed such as exact spatial location in 360° peripheral vision; the number of (and multiple) dots presented; nor correlations with perceived auditory spatialisation.

A broad examination of research into auditory perception, reveals that rhythm of stimuli and timescale, may be important variables. Research has investigated the duration of the stimulus gap (between auditory and visual stimulus) and its length before illusory fragmentation (Shams, 2002). Whilst this research explores the elementary formations of rhythm, research has not investigated the gap size that would cause perceptual fusion between the two stimuli; nor has it investigated the combination of various durations to form rhythmic motifs, and the potential for auditory rhythm motifs transpose to create visual rhythms. Shams (2005) did note that
pilot research…

> *"confirmed that whether beeps and flashes were presented simultaneously or with slight temporal offset made little difference to behavioural reports of illusory perception"* (Pg 1923).

To have confidence in this important conclusion, there is a need for closer and systematic examination of the spatial disparity and duration of the stimulus gap for both large and short durations and their effect on fusion.

Manipulation of Fine-Grained Time Scale

Time scale is an important variable that has been employed and manipulated in many illusory paradigms. The capture of sensory information is accrued by the provision of timescale; therefore, misperception sometimes occurs from insufficient time to acquire sensory information. Employment of the micro time scales as a variable includes Microsounds (Roads, 1996; 2001) with stimuli generated at 600ms or less; the Octave / Scale / Chromatic Illusion (Deutsch, 1981; 1975) generate auditory stimuli at 250ms durations; The Illusory Continuity of Tones (Bregman, 1999) generates noise at 50ms or less; and the Auditory Driving of Visual Flicker (Shipley, 1964) generates auditory stimuli at 150ms. The micro time scale and limited duration of the auditory stimulus employed by the Sound Induced Illusory Flash is an important variable, as the illusion fragments when the beeps expand in duration from 70ms onwards to 100ms.

The structural variable of rhythm has been explored in some depth in current research on the Sound Induced Illusory Flash, although expansion of this variable would be problematic as the micro time scale of auditory stimulus presentation is too short to accurately perceive rhythm. Further, participants would require musical knowledge of simple rhythms to articulate the visual rhythms perceived.

Overall, significant gaps exist in the research of structural variables that may affect the Sound Induced Illusory Flash. Further exploration of the illusion is necessary to provide a greater understanding of the illusion and auditory-cross modal integration. More importantly, and for the present context, it offers potential for creative applications, and new multimedia techniques.

One way to explore the illusion is to examine other possible variables that have been manipulated in the generation of other uni-modal and cross-modal illusions that might be applied to the Sound Induced Illusory Flash.

Frequency and Pitch interval

Frequency as a stimulus variable (as apposed to the sensory dominant and intensively researched rhythm) for cross-modal manipulation was introduced by Marks' exploration on the Mediation of Brightness, Pitch, and Loudness (1974). This psychophysical research does not exhibit cross-modal manipulation, but cross-modal associations between auditory and visual stimulus (greyscale hue and pitch). However, the employment of pitch as a variable has been effective in uni-modal auditory illusions.

The Octave Illusion (Deutsch, 1974) (Deutsch, 1983), Scale Illusion (Deutsch, 1975) and the Chromatic illusion (Deutsch, 1987), pits the perceptual grouping principles of similarity of frequency and spatialisation against one another resulting in an illusory percept based on pitch proximity. Both of these variables – pitch proximity and spatial location – may translate to cross-modal illusions where the variables influence the judgement of visual stimuli and their effect on visual perception assessed.

Harmonic Relationship

The variable of intervallic close harmonic relationship is used to distort motion direction of a consecutive pitch. This variable is most notably exploited by the Tritone Paradox (Deutsch, 1986) that employs multiple layered frequencies at the octave; the Shepard Tone (Shepard, 1964) (Risset, 1972) (Risset, 1986) employs multiple layered frequencies at the octave or the augmented 5th that is further combined with variables of phasing and auditory fixation points.

There is a significant gap in current research on the application of pitch interval and spatialisation in cross-modal auditory visual perception. This gap is reflected in studies of the Sound Induced Illusory Flash that only employs the variables of micro time scales with allusions to rhythm, to distort visual temporal perception, with no investigation into pitch interval and spatialisation variables.

The vast majority of auditory illusions use rhythm and micro time scales as variables to manipulate perception. However, pitch interval and spatialisation were used for purposes of articulation, the generation of perceived motion and spatialisation, therefore there application as variables to articulate visual rhythm, create visual motion and spatialisation should translate to the Sound Induced Illusory Flash.

Experiment Design

Hypothesis

It is hypothesised that the application of pitch interval to the Sound Induced Illusory Flash generates two auditory fixation points corresponding to the high and low pitches (reflective of the Octave illusion), articulating the apparent dot to flicker accordingly and rhythmically, whilst emphasising the illusory effect.

An interval of an octave is employed, as this interval is the closest harmonically to the unison, and therefore the most conservative option for manipulation of the variable.

The application of the variable of transmission from monaural to binaural presentation of the auditory stimulus is hypothesised to further emphasise the two auditory fixation points corresponding to the high and low pitches, articulating the dot to flicker accordingly and emphasising the illusory effect of a single dot. When applied to multiple dots, it is hypothesised that the pitch associated dots will localise according to the spatialisation of the auditory stimuli.

Method

Participants

A sample of 40 participants naïve to the illusion were recruited. They were Psychology 1A students from the University of Western Sydney and received course credit for their participation. Participants were aged between 17 and 54 years (M = 21.175, SD = 6.68), with more female participants than male participants (36 female, 4 male). People reporting a hearing impairment, visual impairment (corrected to normal vision), severe migraines or epilepsy were excluded from testing.

Stimuli

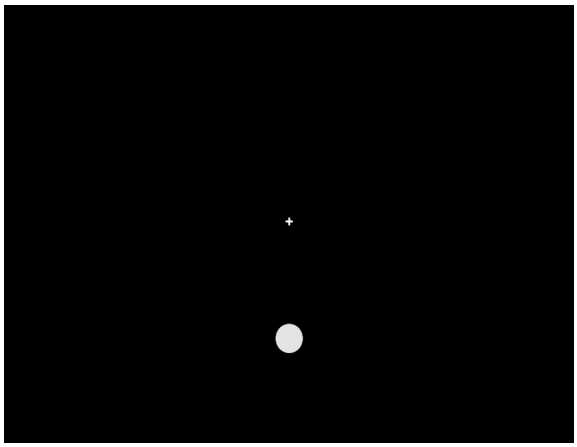


Figure 2. *Screen Capture of the visual stimulus. The fixation point is the centrally located cross, with the dot presented.*

The visual stimulus consisted of a centrally located fixation point, and a single white dot positioned below the centre of the screen that is located in the participants' peripheral vision. In Illusory trials the dot was presented 23ms after the auditory stimulus onset and for the duration of 17ms. The visual stimulus remained the same in the trials, with the independent variables concerning only the aural stimuli.

The auditory stimulus consisted of a sine tone generated every 50ms and lasted for a total of 7ms (attack 2ms, sustain 3ms, decay 2ms).

The variables manipulated were:

- The number of auditory stimulus beeps, presented at two, three, four and five generations.
- Intervallic frequency with the Unison (261Hz) versus the Octave (261 and 523Hz).
- Transmission of the sound through headphones with monaural versus binaural.

The interval variable (unison versus octave) was presented within subjects and the transmission variable (monaural versus binaural) between subjects.

Beeps	Unison Interval		Octave Interval	
Presented	Monaural	Binaural	Monaural	Binaural
2	X	X	X	X
3	X	X	X	X
4	X	X	X	X
5		X	X	X
	X			

Table 1. Experiment Conditions *Table.*

Yielding 16 conditions in total, each condition was presented 6 times. Catch trials were presented with equal chance of the stimulus being illusory or not. The catch trials consisted of the presentation of physical flashes equivalent to the number of auditory beeps, eg. two physical flashes with two auditory beeps; three physical flashes with three auditory beeps; four physical flashes with four auditory beeps; and five physical flashes with five auditory beeps. In catch trials, the dot was presented every 50ms and simultaneous with the auditory stimulus onset.

Equipment

Participants were located at computer workstations with their head positioned on a chin rest 40cm from the computer monitor and eye level with the fixation point. A Mac Pro G5 with a Diamond digital CRT monitor was used with sound transmitted through AKG K601 headphones. MAX / MSP was used to construct an application that generated the auditory and visual stimulus; presented the trials in a randomised and collected order (using the urn object); generated the questionnaire; and collected the participants responses in a text file.

Procedure

Participants were instructed to place their head on the chin rest and focus on the fixation point. The dot would be presented but to use their peripheral vision to count the number of times it was presented.

The task required them to state on a multiple choice questionnaire the number times the dot was presented, ranging from one event to five events. Further, with a time limit of 8 seconds to respond, participants had to state their confidence level of their decision, ranging from not confident, unsure, very confident.

Results

The experiment recovered the illusory effect with results suggesting that pitch interval and binaural transmission enhanced the illusory effect.

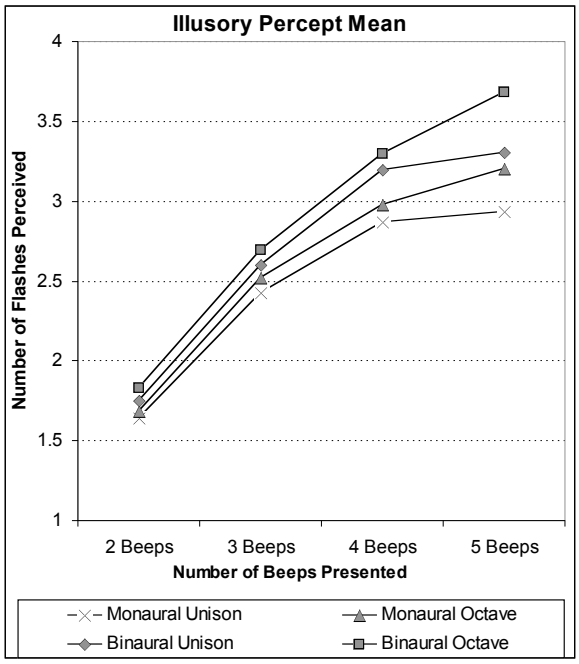


Figure 3. Graph illustrating the means of the perceived number of Flashes.

Participants had a greater performance accuracy mean for two beep presentations, that continued although degraded through to five beep presentations. This degradation of performance accuracy mean was also present in the catch trials, which suggest attentional issues.

Over the two to five beep presentations, the binaural octave condition consistently provided the highest mean, followed by the monaural octave and binaural unison conditions, with the monaural unison condition providing the lowest mean.

These results indicated that modifying the auditory stimulus with the octave interval and binaural transmission enhanced the illusory effect, suggesting that the contrast in auditory stimulus draws attention to, and articulates the auditory rhythm.

References

Alais, David. and Burr, David. 2004. "The Ventriloquist Effect Results from Near-Optimal Bimodal Integration" *Current Biology*. 14: 257 - 262.

Bhattacharya, Joydeep. Shams, Ladan. and Shimojo, Shinsuke. 2002. "Sound-Induced Illusory Fash Perception: Role of Gamma Band Responses" *Cognitive Neuroscience and Neuropsychology*. 13: 14.

Blauert, Jens. 1996. Spatial Hearing - Revised Edition: The Psychophysics of Human Sound Localization. The MIT Press.

Bregman, Albert. 1999. *Auditory Scene Analysis*. Cambridge Massachusetts: The MIT Press.

Chambers, Christopher D. and Mattingley, Jason B. 2002. "The Octave Illusion Revisited: Suppression or Fusion Between Ears" *Journal of Experimental Psychology: Human Perception and Performance*. 28: 6. 1288–1302.

Deutsch, Diana. 1998. *The Psychology of Music*. Second Edition. California: Academic Press.

Deutsch, Diana. 1974. "An Auditory Illusion" *Nature*. 251: 307-309.

Deutsch, Diana. 1975. "Two-channel listening to musical scales" *Journal of the Acoustical Society of America*. 57: 1156-1160.

Deutsch, Diana. 1978. "Delayed Pitch Comparisons and the Principle of Proximity" *Perception and Psychophysics*. 23, 3: 227-230.

Deutsch, Diana. 1981. "The Octave Illusion and Auditory Perceptual Integration" *Hearing Research and Theory*. 1: 99-142.

Deutsch, Diana. 1983. "The octave illusion in relation to handedness and familial handedness background", *Neuropsychologia*. 21: 289-293.

Deutsch, Diana. 1988. "The Semitone Paradox" *Music Perception*. 6, 2: 115-132.

Deutsch, Diana. and Roll, Philip L. 1976. "Seperate "What" and "Where" Decision Mechanisms in Processing a Dichotic Tonal Sequence" *Journal of Experimental Psychology: Human Perception and Performance*. 2, 1: 23-29.

Marks, Lawrence E. 1974. "On the Associations of Light and Sound: The Mediation of Brightness, Pitch, and Loudness", *American Journal of Psychology*. Vol 87: 173-188.

McGurk, Harry. and Macdonald, John. 1976. "Hearing lips and seeing voices" *Nature*. 264: 746 – 748.

Müller-Lyer, F.C. 1889. "Optiche Urteilstäuschungen", *Dubois-Reymonds Archiv für Physiologie*. 7: 263–270.

Risset, Jean-Claude. 1972. *Musical Acoustics*. Paris: IRCAM.

Risset, Jean-Claude. 1986. "Pitch and rhythm paradoxes: Comments on 'Auditory paradox based on fractal waveform'" *The journal of the Acoustical Society of America*. 80, 3:961-962.

Roads, Curtis. 1996. *The Computer Music Tutorial*. Cambridge, Massachusetts: The MIT Press.

Roads, Curtis. 2001. *Microsound*. Cambridge, Massachusetts: The MIT Press. 86-106, 330-351.

Shams, Ladan. 2002. "Integration in the Brain - The Subconscious Alteration of Visual Perception by Cross-Modal Integration" *Science and Consciousness Review*. 1: 1-4.

Shams, Ladan. 2005. "Sound induced flash illusion as an optimal percept" *Neuroreport*. vol 16: 17.

Shams, Ladan. Iwaki, Sunao. Chawla, Aman. and Bhattacharya, Joydeep. 2005. "Early Modulation of Visual Cortex by Sound: an MEG Study" *Neuroscience Letters*. 378: 76–81.

Shams, Ladan. Kamitani, Yukiyasu. and Shimojo, Shinsuke. 2002. "Visual Illusion Induced by Sound" *Cognitive Brain Research*. 14: 147-152.

Shams, Ladan. Kamitani, Yukiyasu. Thompson, Samuel. and Shimojo, Shinsuke. 2001. "Sound Alters Visual Evoked Potentials in Humans" *Cognitive Neuroscience and Neuropsychology*. 12, 17: 3849-3852.

Shepard, Roger. 1964. "Circularity in Judgements of Relative Pitch" *The journal of the Acoustical Society of America*. 36, 12: 2346-2353.

Shimojo, Shinsuke. Scheier, Christian.Nijhawan, Romi. Shams, Ladan. Kamitani, Yukiyasu. and Watanabe, Katsumi. 2001. "Beyond Perceptual Modality: Auditory Effects on Visual Perception" *Acoustic Science and Technology*. 22, 2: 61-67.

Shimojo, Shinsuke. and Shams, Ladan. 2001. "Sensory Modalities Are Not Separate Modalities: Plasticity and Interactions" *Current Opinion in Neurobiology*. 11: 505-509.

Shipley, T. 1964. "Auditory Flutter Driving of Visual Flicker" *Science*. vol 145: 1328-1330.